

Assignment 2 - Machine Learning in Computational Biology
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Interpretable Machine Learning for
Breast Cancer Classification: A
Repeated Nested Cross-Validation
Approach

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Abstract

This project focuses on building a machine learning classifier to help distinguish between malignant and benign breast tumors, based on data from fine needle aspirates (FNA) of breast tissue. The dataset includes 512 samples, each with 30 numerical features describing characteristics of cell nuclei. To choose the best model while avoiding overfitting and ensuring generalizability, we implemented a fully object-oriented repeated nested cross-validation (rnCV) pipeline. Six algorithms were tested: Logistic Regression (LR), Gaussian Naive Bayes, Linear Discriminant Analysis, Support Vector Machines, Random Forests, and LightGBM. Given that the dataset is slightly imbalanced and includes correlated features, we compared models using several evaluation metrics, mainly Matthews Correlation Coefficient (MCC), F1 score, and AUC. Logistic Regression had the best overall performance with a median MCC of 0.938, F1 score of 0.961, and AUC of 0.996. Although Support Vector Machines performed similarly, LR was selected as the final model due to its interpretability. The top coefficients in the LR model highlighted features like area and texture as important for classification. This project shows how interpretable simple models can still perform very well, which is useful when trying to connect machine learning results to biological understanding. Code and processing pipeline are available at: https://github.com/vashqu/Assignment-2/tree/vchatzit_solution.

Introduction

Breast cancer is one of the most common and life-threatening diseases among women worldwide, being the most common type of cancer and a type of cancer where half of the cases are not related to risk factors[1]. Early and accurate diagnosis plays a critical role in improving treatment outcomes and reducing unnecessary medical procedures. Fine needle aspiration (FNA) is a minimally invasive method for examining breast tissue, and the resulting cytological data can be used to differentiate between benign and malignant tumors. However, interpreting this data manually can be challenging and subjective.

In this project, we focus on developing a machine learning pipeline that can classify tumor samples as benign or malignant based on 30 numerical features derived from FNA images. These features describe the physical characteristics of cell nuclei, such as radius, texture, area, and concavity, and are provided for each sample in three forms: mean, standard error (SE), and worst value.

Studies have explored the use of machine learning models on breast cancer datasets, including the well-known Wisconsin Breast Cancer dataset. Research has shown that algorithms such as Support Vector Machines and Random Forests can achieve high accuracy on this type of data[2]. However, interpretability remains a major concern in clinical settings, where understanding the reasoning behind a model's prediction is just as important as performance. Logistic Regression is commonly used in healthcare-related machine learning due to its transparency, as model coefficients can be directly mapped to feature importance[3]. This aligns with the growing emphasis on explainable AI (XAI) in biomedical applications.

Our main objective was to evaluate and compare the performance of six classification algorithms: Logistic Regression (LR), Gaussian Naive Bayes (GNB), Linear Discriminant Analysis (LDA), Support Vector Machines (SVM), Random Forests (RF), and LightGBM. We implemented a

fully object-oriented repeated nested cross-validation (rnCV) pipeline to systematically tune and evaluate each model. Given the class imbalance in the dataset and the importance of interpretability in medical applications, we focused on metrics such as the Matthews Correlation Coefficient (MCC), F1 score, and Area Under the Curve (AUC) to guide model selection.

Our contributions include the implementation of a reusable rnCV pipeline, a comparison of multiple classifiers on medical data, and an analysis of model interpretability using logistic regression coefficients.

Materials and Methods

Dataset Description

The dataset consists of 512 samples collected from fine needle aspirates (FNA) of breast masses. Each sample includes 30 numerical features derived from image analysis of cell nuclei. These features quantify ten biological properties: *radius*, *texture*, *perimeter*, *area*, *smoothness*, *compactness*, *concavity*, *concave points*, *symmetry*, and *fractal dimension*. Each of these properties is described in three forms: *mean*, *standard error*, and *worst* (i.e., the largest values), leading to a total of 30 features per sample.

The target variable, **diagnosis**, is binary, indicating whether the tumor is benign (B) or malignant (M). This classification task poses a challenge due to slight class imbalance and potential multicollinearity between features.

Exploratory Data Analysis

We began by checking for missing values, data types, and class distribution. The dataset was clean with no missing entries. Class distribution showed a slight imbalance favoring benign cases, with 321 benign and 191 malignant samples, suggesting the need for stratified cross validation to respect the class distribution. We also checked for duplicates with none present. Boxplots and heatmaps were used to understand feature distributions and relationships. Most correlations fall below 0.2 with very few reaching above 0.3, suggesting limited linear dependency between the features and no strong block structures. Since no strong dependencies were observed we inferred that the missing values are likely Missing Completely at Random and decided to handle them by imputing the median as it is also robust to outliers given our data is mostly right-skewed. Later on, a Principal Component Analysis (PCA) was performed to visualize data separability, showing reasonable distinction between benign and malignant samples in reduced dimensions, with the benign forming a tight cluster.

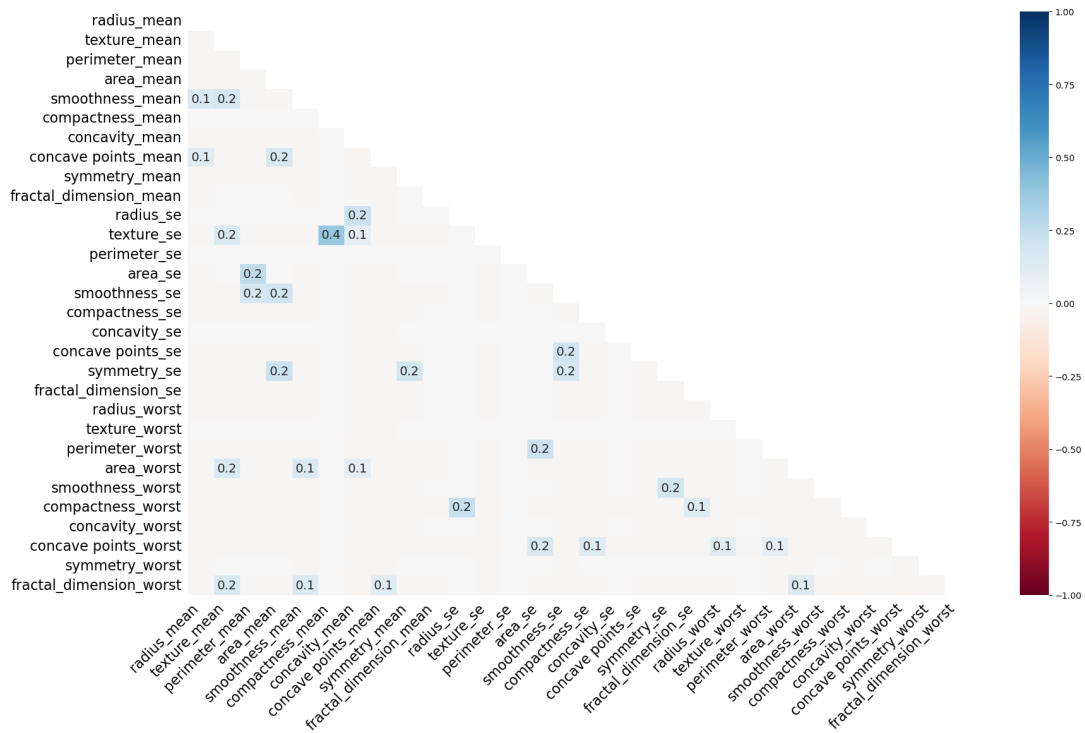


Figure 1: Feature correlation heatmap showing pairwise relationships among the 30 features. Strong positive correlations are visible among size-related features (e.g., radius, perimeter, and area).

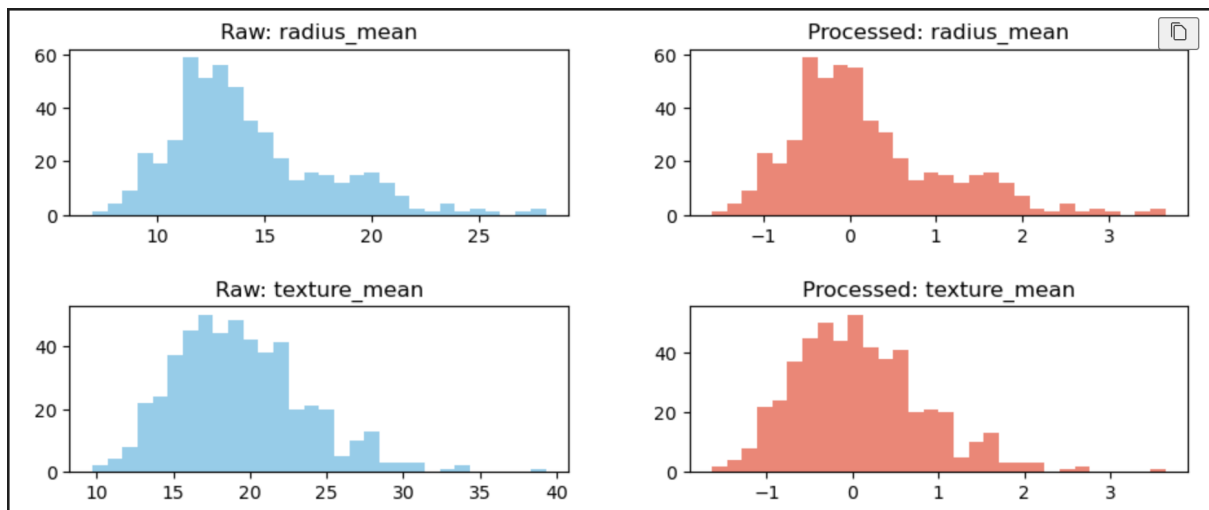


Figure 2: Distributions of raw (left) and standardized (right) values for two features. Standardization brings different features to a similar scale, which helps many models perform better.

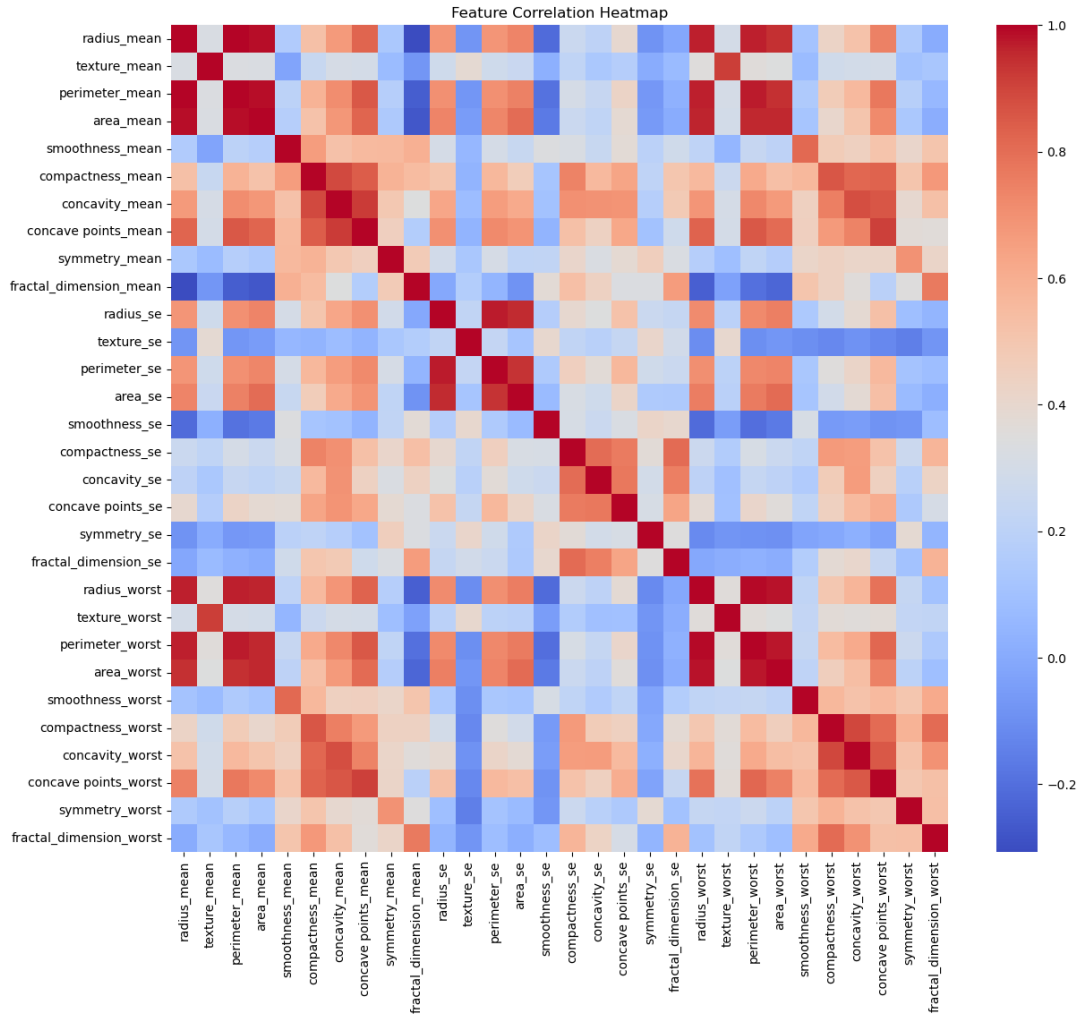


Figure 3: Detailed correlation matrix (full view) visualizing redundancy and multicollinearity among features.

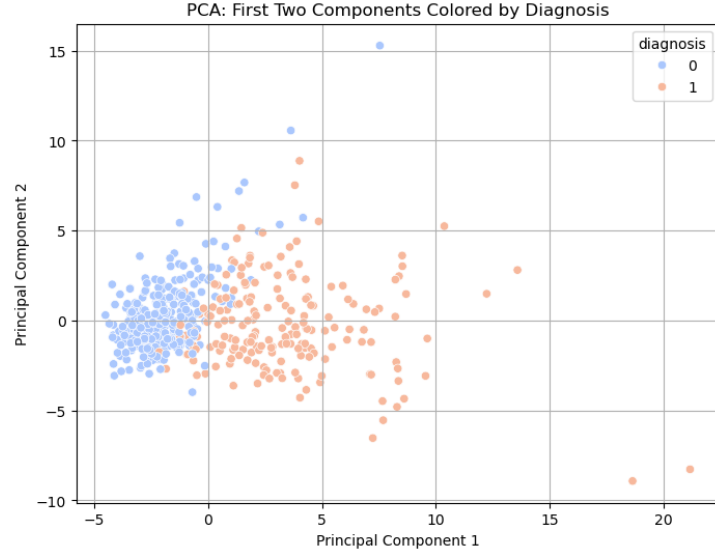


Figure 4: PCA plot of the first two principal components, colored by diagnosis. Reasonable separation between benign and malignant samples is observed in the reduced feature space.

Before training, all features were scaled using RobustScaler to ensure that models were not biased by different feature magnitudes. Preprocessing was integrated directly into the machine learning pipeline to prevent data leakage.

Repeated Nested Cross-Validation (rnCV) Pipeline

We implemented a reusable Python class that performs repeated nested cross-validation, structured using object-oriented programming principles. The outer loop (5-fold) estimates generalization performance, while the inner loop (3-fold) is used for hyperparameter tuning. This process was repeated 10 times to ensure robust performance estimates. The pipeline accepts a list of classifiers and corresponding hyperparameter grids. In each repetition, models were trained and tuned within the inner loop using multiple scoring metrics (e.g., MCC, F1, AUC), and the best-performing model was evaluated on the outer test fold. Scores across the 50 outer-fold evaluations (5 folds $\times$ 10 repetitions) were aggregated to compare algorithm performance. To account for class imbalance, stratified folds were used, and scoring metrics were chosen to reflect the clinical significance of false positives and false negatives.

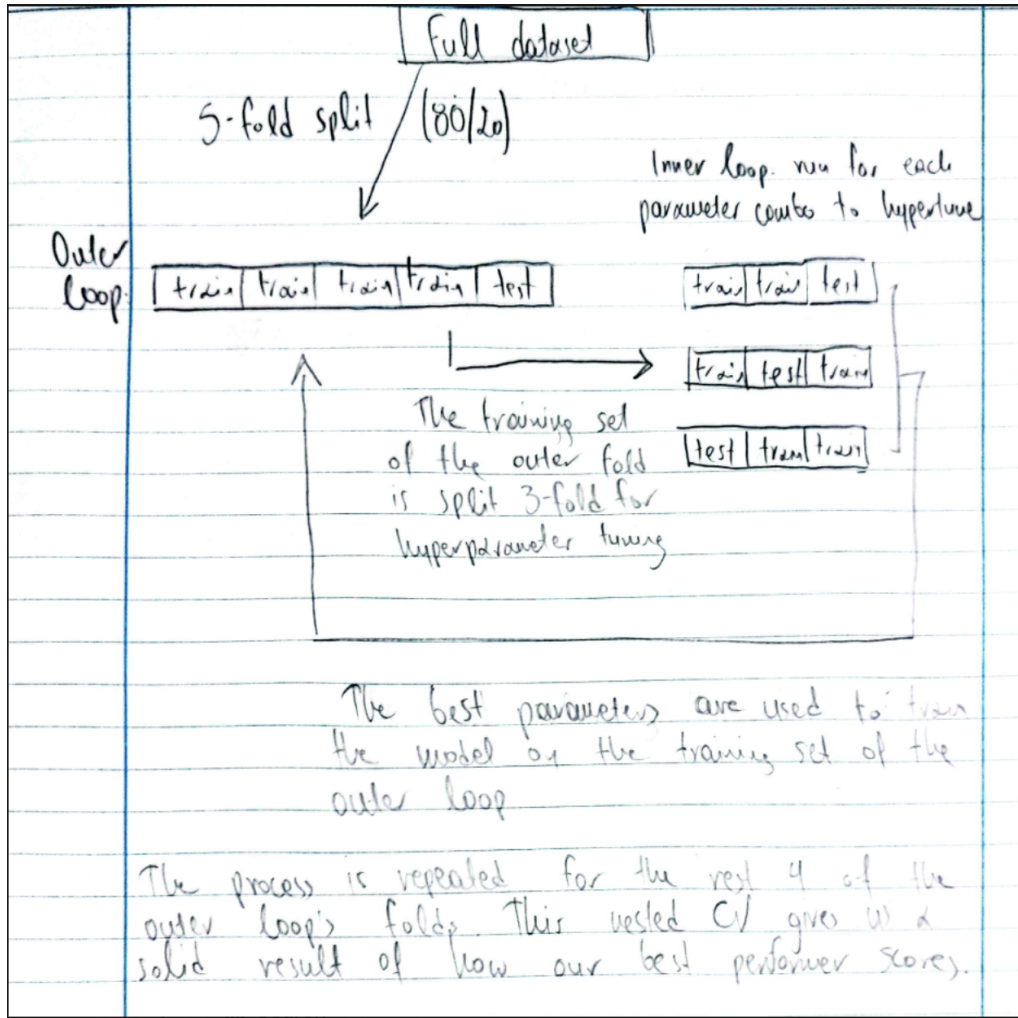


Figure 5: Hand-drawn schematic of the repeated nested cross-validation (rnCV) process. The outer loop splits the data into 5 folds for performance estimation, and the inner loop applies 3-fold cross-validation on the training portion of each outer fold for hyperparameter tuning. This design ensures robust model selection while preventing data leakage.

Model Selection and Evaluation Metrics

We evaluated each model using several metrics. The *Matthews Correlation Coefficient (MCC)* was used to assess balanced performance, *F1 Score* was selected to balance precision and recall, and *Area Under the Curve (AUC)* was used to measure the ranking quality of predictions. These metrics were chosen because they are suitable for imbalanced datasets and align with the priorities of medical decision-making tasks. The final model selection was based on the median values of these metrics across all outer loop evaluations. In cases where models had similar performance, interpretability was used as a deciding factor.

Results and Discussion

Model Comparison and Evaluation

We ran a repeated nested cross-validation (rnCV) procedure using 10 rounds of 5 outer folds and 3 inner folds. Each model was tuned in the inner loop and evaluated on the corresponding outer loop test fold. The median scores across the 50 outer test folds for each model are summarized in Table 1.

Table 1: Median performance scores across 50 outer test folds.

Model	MCC (Median)	AUC (Median)	F1 Score (Median)
Logistic Regression (LR)	0.938	0.996	0.961
Support Vector Machine (SVM)	0.935	0.995	0.959
Linear Discriminant Analysis (LDA)	0.924	0.990	0.950
Random Forest (RF)	0.920	0.986	0.946
LightGBM	0.915	0.984	0.943
Gaussian Naive Bayes (GNB)	0.881	0.965	0.910

Logistic Regression consistently showed the best or nearly-best performance across all evaluation metrics. Although Support Vector Machines performed similarly, we selected LR as the final model due to its transparency and ease of interpretation in medical applications.

Model Performance and Interpretability

Logistic Regression provided the most interpretable model. By inspecting the top positive and negative coefficients, we identified key features influencing classification decisions. Among the strongest positive predictors for malignancy were *texture (worst)*, *area (standard error)*, and *area (worst)*. These findings align with medical intuition, as larger and more irregular cell structures are commonly associated with malignant growths. The model’s ability to highlight such interpretable features makes it particularly suitable for clinical decision support, where transparency is essential.

We evaluated six classification models using a repeated nested cross-validation (rnCV) approach, with 10 repetitions of 5-by-3-fold cross-validation. The models were compared using three key metrics: *Matthews Correlation Coefficient (MCC)*, *Area Under the Curve (AUC)*, and *F1 Score*. The results for the top three models—Logistic Regression, Support Vector Machines, and LightGBM—are presented along with 95% confidence intervals computed via bootstrapping.

Ranking Strategy and Confidence Intervals

To identify the best-performing models, we ranked them by their median Matthews Correlation Coefficient (MCC), a reliable metric for imbalanced classification. Unlike accuracy or F1 score, MCC accounts for all entries in the confusion matrix, making it especially suitable when class sizes are unequal.

The top-performing models by MCC were:

1. Logistic Regression (ElasticNet) — $\text{MCC} \approx 0.938$
2. Support Vector Machine (SVM) — $\text{MCC} \approx 0.937$
3. LightGBM — $\text{MCC} \approx 0.916$

These results indicate that both regularized linear models and nonlinear kernel-based models perform well on this dataset. We proceeded with bootstrap resampling to compute 95% confidence intervals (CIs), followed by statistical comparison to finalize the best model for deployment.

Figures 6 through 8 show the 95% confidence intervals for MCC, AUC, and F1 score, respectively. Figure 6 shows that MCC confidence intervals for Logistic Regression narrowly outperform SVM, though the intervals do overlap. In Figure 7, AUC scores were consistently high across all models, with LR and SVM nearly tied. Figure 8 shows that LR achieved the highest median F1 score and the tightest CI.

Based on these results, we selected Logistic Regression as the final model. Its performance was on par with SVM, but its interpretability and transparent coefficient-based structure made it more suitable for clinical use.

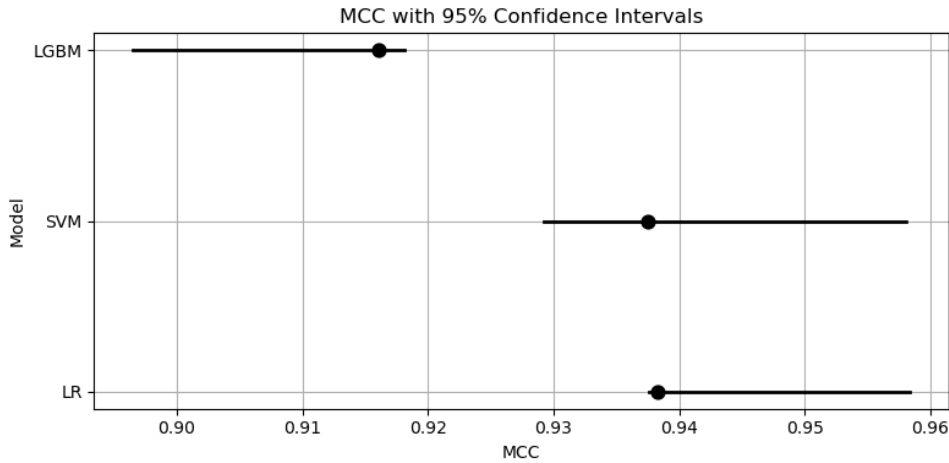


Figure 6: 95% confidence intervals for MCC across the top three models. Logistic Regression (LR) narrowly outperforms SVM, with slightly overlapping intervals.

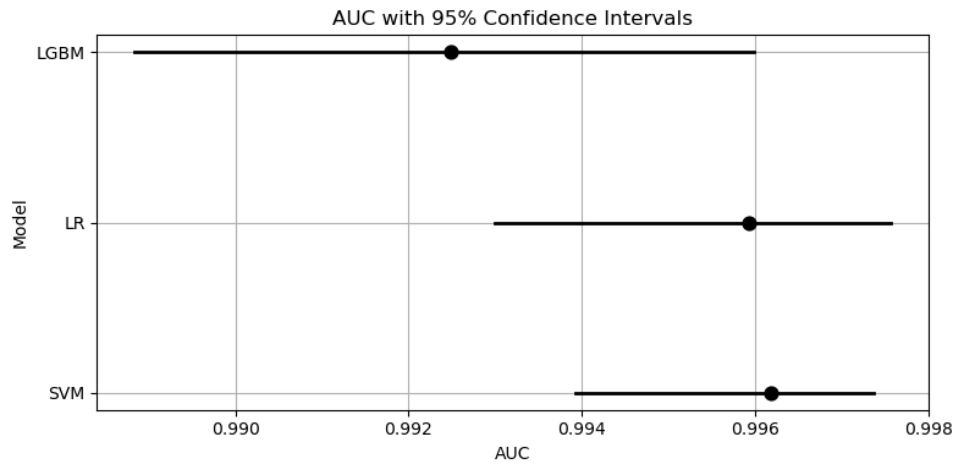


Figure 7: 95% confidence intervals for AUC. All models achieve high AUC values, with LR and SVM showing comparable performance.

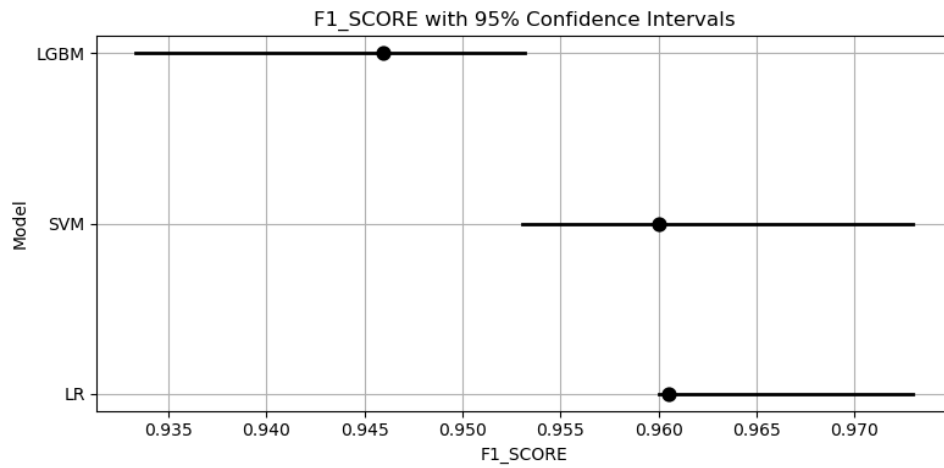


Figure 8: 95% confidence intervals for F1 score. LR again leads with the highest median and tightest confidence interval.

Logistic Regression interpretation

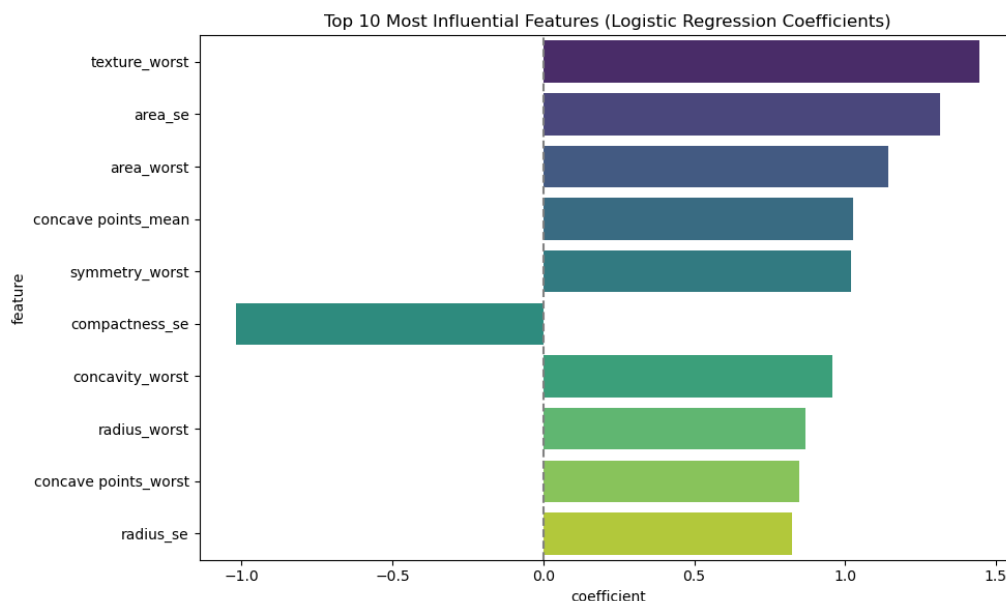


Figure 9: Top 10 most influential features according to the coefficients of the final Logistic Regression model. Positive values contribute to a malignant prediction, while negative values contribute to a benign prediction.

We examined the coefficients of the final trained Logistic Regression model to identify which features most strongly influenced its predictions. The top features are shown in Figure 9. The features with the largest positive coefficients were `texture_worst`, `area_se`, and `area_worst`. High values in these features increased the model’s likelihood of predicting a sample as malignant.

This aligns with medical intuition, as malignant tumors often have more irregular, larger nuclei and greater variation in texture. These findings reinforce the interpretability advantage of Logistic Regression, as the coefficients directly reflect the direction and magnitude of influence for each feature. On the other hand, features like `compactness_se` had a negative coefficient, indicating that lower values may be more associated with malignant classifications in this dataset.

Conclusion

In this project, we developed and evaluated a machine learning pipeline to classify breast tumor samples as benign or malignant using numerical features derived from fine needle aspirates (FNA). The dataset included 30 features per sample, and we implemented a fully object-oriented repeated nested cross-validation (rnCV) framework to ensure robust model selection and generalizability.

We compared six classification algorithms using multiple metrics, with a focus on Matthews Correlation Coefficient (MCC) due to its suitability for imbalanced classification problems. Logistic Regression with Elastic Net regularization emerged as the best model overall, with strong median performance across MCC, AUC, and F1 score. Its coefficient-based structure

also allowed for interpretation of key features driving classification decisions—an important factor in clinical applications.

Although Support Vector Machines performed similarly in terms of predictive power, we prioritized interpretability and reproducibility in our final model choice. One limitation of our approach is that we relied only on numerical imaging-derived features. In future work, integrating additional clinical metadata—such as patient age, family history, or hormonal receptor status—could improve classification accuracy and allow for more personalized predictions. External validation on independent datasets and exploration of explainability techniques for non-linear models also represent valuable directions for extension.

Overall, the project demonstrates how interpretable machine learning models can achieve high performance while offering meaningful insights into biomedical data.

Author Notes

This report was prepared with the assistance of generative AI tools, including **ChatGPT-4 (OpenAI)**, **Grok 3 Beta (xAI)**, and **DeepSeek**. These tools were used to support tasks such as proofreading, scientific writing, code debugging and L^AT_EX formatting. All analytical decisions, interpretations, and conclusions were independently made and critically reviewed by the author.

References

References

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